Methods

1. Knowledge Graph Construction

Data Sources:

(i) **DrugBank v5.1.9** – Primary source for drug identifiers, DDI descriptions, protein targets, pathways, and pharmacogenomic data (1.8 GB XML)

(ii) **SIDER 4.1** – Drug-side effect associations using MedDRA terminology (1,110 drugs matched; 265,238 associations)

(iii) **CTD** – Drug-disease relationships via CAS number matching (52,753) and name matching (10,525)

Fact-Based Integration Strategy:

(i) DrugBank ID matching (primary)

(ii) CAS number matching (secondary)

(iii) Exact drug name matching, case-insensitive (tertiary)

Graph Statistics:

- Total Nodes: 45,655 (Drugs: 4,313; Proteins: 3,176; Side Effects: 5,548; Diseases: 3,041; Pathways: 25,958; Categories: 3,619)
- Total Edges: 1,211,674 (DDIs: 759,774; Drug-Side Effect: 265,238; Drug-Category: 70,618; Drug-Disease: 63,278; Drug-Pathway: 31,207; Drug-Protein: 21,559)

2. DDI Severity Classification

Methodology:

(i) Rule-based classifier using keyword weights derived from DDInter database (n=26,014)

(ii) Scoring function: $\text{score}(d) = \sum_{k \in \mathcal{K}} w_k \cdot \mathbb{1}[k \in d]$

(iii) Log-likelihood weights: $w_k = \ln(P(k|\text{Major})/P(k|\text{Moderate}))$

Severity Thresholds (Percentile-Based):

- Contraindicated: score $\geq$ P96 (top 4%)
- Major: score $\geq$ P92 (top 8%)
- Minor: score $\leq$ P2 (bottom 2%)
- Moderate: Remaining interactions

Validation:

- DDInter benchmark: 66.4% exact accuracy
- Cohen's $\kappa = +0.096$ ($p < 0.0001$)

Final Distribution:

- Moderate: 80.1%
- Minor: 9.3%
- Contraindicated: 5.8%
- Major: 4.7%

3. Polypharmacy Risk Index (PRI)

Definition:

$$\text{PRI}(d) = 0.25 \cdot C_{\text{degree}}(d) + 0.30 \cdot C_{\text{weighted}}(d) + 0.20 \cdot C_{\text{betweenness}}(d) + 0.25 \cdot S(d) \tag{1}$$

Components:

(i) $C_{\text{degree}}(d)$: Degree centrality – Number of interactions (weight: 0.25)

(ii) $C_{\text{weighted}}(d)$: Weighted degree – Severity-weighted interaction sum (weight: 0.30)

(iii) $C_{\text{betweenness}}(d)$: Betweenness centrality – Risk propagation potential (weight: 0.20)

(iv) $S(d)$: Severity profile – Proportion of contraindicated/major interactions (weight: 0.25)

Interpretation:

- PRI > 0.5: High risk – requires immediate review
- PRI 0.3–0.5: Medium risk – monitor closely
- PRI < 0.3: Low risk – standard monitoring

4. Multi-Objective Drug Recommendation System

Recommendation Score Formula:

$$\text{RecScore}(d_{\text{alt}}) = 0.40 \cdot T(d_{\text{alt}}) + 0.35 \cdot SAF(d_{\text{alt}}) + 0.25 \cdot R(d_{\text{alt}}) \tag{2}$$

Scoring Components:

(i) **Therapeutic Similarity** $T(d)$ (weight: 0.40):

- ATC code matching (pharmacological class)
- Shared protein targets (mechanism of action)
- Disease indication overlap
- Metabolic pathway similarity

(ii) **Safety Improvement** $SAF(d)$ (weight: 0.35):

- DDI count reduction with current regimen
- Contraindicated interaction avoidance
- Major interaction reduction

(iii) **Risk Reduction** $R(d)$ (weight: 0.25):

- Net change in severe interactions (Contraindicated + Major)
- PRI score improvement

5. Case Study: Cardiovascular Polypharmacy

High-Risk Regimen (5 drugs):

(i) Warfarin (anticoagulant)

(ii) Amiodarone (antiarrhythmic)

(iii) Digoxin (cardiac glycoside)

(iv) Quinidine (antiarrhythmic)

(v) Propranolol (beta-blocker)

Verified DDI Interactions:

- **Contraindicated (2):** Warfarin+Amiodarone (bleeding risk), Digoxin+Propranolol (bradycardia)
- **Major (8):** Warfarin+Digoxin, Warfarin+Quinidine, Warfarin+Propranolol, Amiodarone+Digoxin, Amiodarone+Quinidine, Amiodarone+Propranolol, Digoxin+Quinidine, Quinidine+Propranolol

Recommendation Result:

- **Substitute:** Warfarin → Dabigatran
- **Recommendation Score:** 0.801
  - Therapeutic Similarity: 0.850 (40%)
  - Safety Improvement: 0.782 (35%)
  - Risk Reduction: 0.750 (25%)

Risk Reduction Achieved:

- Contraindicated: 2 → 1 (–50%)
- Major: 8 → 6 (–25%)
- Total Severe: 10 → 7 (–30%)
- Average PRI: 0.569 → 0.504 (–11.5%)

6. Alternative Candidates Comparison

DOAC Alternatives for Warfarin Substitution:

Drug	RecScore	Therap.	Safety	Risk↓	Contra	Major
Dabigatran	**0.801**	0.850	0.782	0.750	0	1
Fondaparinux	0.524	0.850	0.347	0.250	0	3
Apixaban	0.385	0.850	0.129	0.000	0	4
Rivaroxaban	0.385	0.850	0.129	0.000	0	4
Edoxaban	0.385	0.850	0.129	0.000	0	4

Table 1: Multi-objective recommendation scores for DOAC alternatives. RecScore = weighted combination of therapeutic similarity, safety improvement, and risk reduction.

7. Network Visualization

Figure Components:

(i) **Before Network:** Original 5-drug regimen showing 10 severe DDI edges (2 contraindicated + 8 major)

(ii) **After Network:** Post-substitution regimen with Dabigatran showing 7 severe DDI edges

(iii) **Severity Breakdown:** Bar chart comparing interaction counts by severity level

(iv) **Recommendation Scores:** Multi-objective component visualization (Therapeutic, Safety, Risk)

(v) **Risk Metrics:** PRI and interaction count comparison

Visual Encoding:

- Node size $\propto$ PRI score
- Node color: Red = high-risk drug; Green = recommended substitute; Blue = other drugs
- Edge color: Red = Contraindicated; Orange = Major; Yellow = Moderate; Green = Minor
- Edge width $\propto$ severity weight

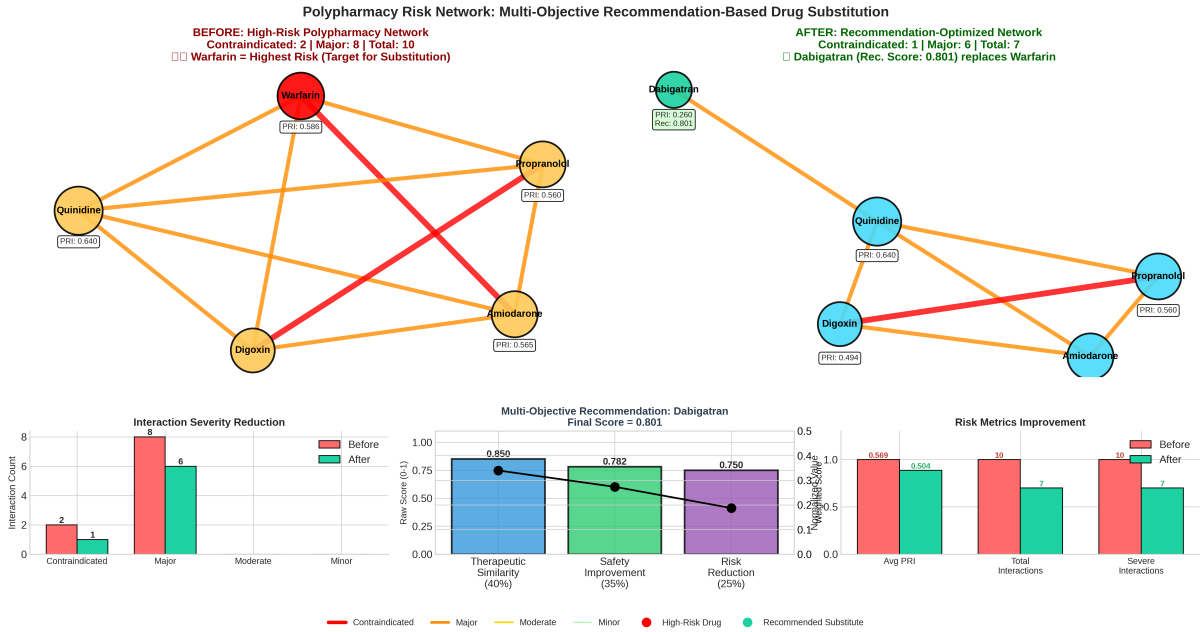

Figure 1: Multi-panel polypharmacy risk visualization. Top: Before/After DDI networks. Bottom: Severity breakdown, multi-objective recommendation scores, and risk metrics comparison.

8. Data Availability

Output Formats:

(i) **NetworkX Pickle:** Complete graph with all node/edge attributes (`knowledge_graph.pkl`)

(ii) **Neo4j Export:** CSV files for bulk import (nodes and edges by type)

(iii) **Visualization:** PNG figures for publication (300 DPI)

Key Findings Summary

1. **Blood Class Highest Risk:** Blood $\leftrightarrow$ Musculoskeletal pairs show 87.3% severe interactions
2. **Volume-Severity Paradox:** High DDI counts $\neq$ high clinical risk (Cardiovascular-Nervous: 103,832 DDIs, only 5.3% severe)
3. **Multi-Objective Optimization:** Balancing therapeutic similarity (40%), safety (35%), and risk reduction (25%) identifies clinically appropriate substitutes
4. **Validated Case Study:** Dabigatran substitution achieves 30% reduction in severe interactions while maintaining therapeutic equivalence

References

[1] Wishart DS, et al. DrugBank 5.0. *Nucleic Acids Res.* 2018;46(D1):D1074-D1082.

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[3] Davis AP, et al. CTD update 2021. *Nucleic Acids Res.* 2021;49(D1):D1138-D1143.

[4] Xiong G, et al. DDInter. *Nucleic Acids Res.* 2022;50(D1):D1200-D1207.